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Data Article

A comprehensive dental dataset of six classes for deep learning based object detection study

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ABSTRACT

This article presents a dental dataset for the improvement of research on deep learning-based detection and classification of dental diseases. The dataset is consisted of 232 panoramic dental radiographs, categorized into six major classes: healthy teeth, caries, impacted teeth, infections, fractured teeth, and broken-down crowns/roots (BDC/BDR). The images were collected from three renowned private clinics in Dhaka, Bangladesh, with the help of an experienced dental practitioner who ensured the confidentiality of patients and high-quality data acquisition using a 64-megapixel Android phone camera. To enhance the value of the dataset for machine and deep learning applications, we applied Contrast-Limited Adaptive Histogram Equalization (CLAHE) for image enhancement and augmented the data. The images were annotated using the CVAT tool and reviewed by dental experts. This benchmark dataset is publicly available and provides a valuable resource for researchers in artificial intelligence, computer science, and dental informatics to promote interdisciplinary collaboration and the development of advanced algorithms for dental disease detection.

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Specifications Table

Subject	Artificial Intelligence, Computer Science, Dental Informatics.
Specific subject area	Machine Learning and Deep Learning based object detection, segmentation, classification of dental diseases.
Type of data	Images, JSON format
Data collection	Dental OPG X-rays were collected from three different private clinics with the help of an experienced dental practitioner. We ensured that all precautions and safety measures were followed. We took special care to keep patient identities confidential. The images were captured using an Android phone's high-resolution camera. The aspect ratio of these images is 16:9, and the median image ratio of these images is 1176 × 603 pixels. The images are categorized into six classes: healthy teeth, caries, infection, impacted teeth, fractured teeth, and BDC/BDR. The dataset is organized into two folders for machine and deep learning model training: the object detection dataset folder and the classification dataset folder. The object detection folder contains 232 original and 604 augmented images and labels. The classification folder contains six distinct files for each class. The images are in JPG format, and the labels are in JSON format. We have split the augmented data into train, test, and valid folders. The dataset's total size is 197 MB, and it is available as a zip file for convenient downloading.
Data source location	Three private dental clinics**:** Prescription Point Ltd, LabAid Specialized Hospital, Ibn Sina Diagnostic and Imaging Center. City/Town/Region: Banani and Dhanmondi, Dhaka Country: Bangladesh
Data accessibility	Repository name: Mendeley Data Data identification number: 10.17632/c4hhrkxytw.4 Direct URL to data: https://data.mendeley.com/datasets/c4hhrkxytw/4

1. Value of the Data

- The Dental OPG X-RAY [1] dataset focused on six major dental conditions, including healthy teeth, caries, impacted teeth, infections, fractured teeth, and broken-down crown/root (BDC/BDR), which were captured on radiographs, impacting dental health across different populations. This dataset is publicly available, and researchers can use it directly in machine learning and deep learning algorithms for object detection and classification purposes.
- This dataset included a diverse collection of 232 dental radiographs labelled with six different classes. This variety allows researchers to explore the application of machine learning and deep learning models for multi-class classification in dental health. The diversity of the dataset in patient age and dental conditions enhances its generalizability across different demographic groups, making it a valuable resource for advancing AI-driven dental diagnostics.
- This dataset provides both original and augmented data and offers a robust foundation for training, testing, and improving machine-learning algorithms. This enables researchers to experiment with different models and techniques, contributing to the advancement of AI in dental healthcare and potential automation of dental diagnostic processes.
- The dataset excluded individuals under 10 years of age to focus on adult dental structures, which vary significantly from those of children. This targeted selection enhances the utility of the dataset in studying age-related variations in dental diseases and the effectiveness of AI models across different age groups.
- This dataset fosters collaboration between computer science and dental health experts, promoting interdisciplinary research that bridges the gap between technology and health care. This dataset serves as a valuable resource not only for AI and computer science researchers but also for dental professionals, educators, and students. It can be used in educational settings to train dental practitioners in the identification of various dental diseases.

2. Background

We developed a benchmark dental disease dataset aimed at advancing research and innovation in dental health, particularly focusing on the detection and classification of multi-class dental diseases [7]. By leveraging machine learning, particularly deep learning, datasets offer valuable insights into the fields of dental informatics, computer science, and healthcare. To create this dataset, we performed several processes, such as careful collection, annotation, and classification of dental radiographs [6]. This standardized dataset fosters collaboration, aids in the development of advanced algorithms, and enhances diagnostic accuracy and treatment outcomes in dental care. As a dedicated data resource, it offers researchers a dependable benchmark for validating algorithms, contributing to the broader knowledge base in the field and supporting both current and future research in dental disease identification and management.

3. Data Description

3.1. Description of major dental disease and symptom

The dataset included six dental conditions. Each condition presents unique symptoms. If it left untreated, can lead to further complications. Accurate diagnosis and early treatment are crucial for the effective management of these dental issues.

3.1.1. Healthy teeth

Healthy teeth are the ideal state of dental health, in which the teeth and surrounding structures are free from damage or disease. Healthy teeth have intact enamel, no signs of decay or infection, and proper alignment and spacing [8]. Healthy teeth serve as a reference point in dental diagnostics, helping differentiate normal anatomy from pathological conditions. In this dataset, images of healthy teeth are crucial for training AI models to recognize what a normal tooth looks like, ensuring the accurate identification of abnormalities.

3.1.2. Caries

Tooth decay or cavities are known as caries and are one of the most common dental problems. Caries occur when acids produced by bacteria in the mouth erode the tooth enamel, leading to the formation of cavities [9]. Without treatment, caries may spread to deeper layers of the tooth, resulting in pain, infection, and risk of tooth replacement. Radiographic images of caries usually display darkish regions on the tooth surface or within the tooth structure, indicating decay. The early detection and diagnosis of caries are crucial for preventing additional damage and protecting oral health.

3.1.3. Impacted teeth

Impacted teeth are those that are blocked from erupting properly into the mouth. This problem frequently occurs in wisdom teeth, which frequently lack sufficient space to grow or are displaced. Impacted teeth can result in discomfort, pain, and swelling and have the potential to create additional dental complications, such as infection or damage to nearby teeth. Radiographs of impacted teeth typically reveal the tooth being entrapped under the gum or being situated in an abnormal position within the jawbone. Early detection is important to avoid complications and determine the necessity of surgical treatment.

3.1.4. Infection

Dental infections, such as abscesses, are dangerous conditions in which bacteria infiltrate the inner layers of the tooth or the surrounding tissues, resulting in inflammation and pus accumulation. These infections can lead to intense pain, swelling, and fever and may spread to other parts of the body if not treated early. Dental infections often appear as dark spots surrounding

Table 1
Attributes of the dataset.

Serial No	Classes	Number of images	Gender	Age range
1	Healthy Teeth	223	Male & Female	10 to 90 years old
2	Caries	119	Male & Female	10 to 90 years old
3	Impacted Teeth	87	Male & Female	10 to 85 years old
4	BDC/BDR	52	Male & Female	16 to 90 years old
5	Infection	23	Male & Female	16 to 85 years old
6	Fractured Teeth	13	Male & Female	16 to 85 years old

tooth roots on radiographs, indicating either bone deterioration or infection [8]. Accurate diagnosis and timely treatment are crucial to prevent the spread of infection and to preserve the affected tooth.

3.1.5. Fractured teeth

Fractured teeth refer to any break or crack in a tooth that can occur due to trauma, biting on hard objects, or weakened tooth structure. Fractures can vary in severity from minor cracks in the enamel to deep cracks that extend into the tooth pulp or root. These fractures can cause pain, sensitivity, and increase the risk of infection. Radiographs of fractured teeth help identify the extent of the damage and guide the choice of treatment, which may include bonding, crowns, or extraction.

3.1.6. Broken-down crown/root (BDC/BDR)

Broken-down crowns or roots refer to teeth that have undergone significant structural damage, often to the point where they cannot be restored using simple dental procedures. This syndrome can arise from significant dental deterioration, numerous dental procedures, or serious physical injuries. Teeth that have broken down crowns or roots may be in a severely damaged state, with significant portions missing or decayed. Radiographs of these teeth commonly show significant dental caries, fractures, or other indications of structural deterioration [10]. Sometimes, damaged teeth require extraction or complex restoration activities to repair the damage. Fig. 1 shows the images of each class.

3.2. Description of the dataset folders

Fig. 2 depicts the folder structure of the datasets, and the process of the dataset preparation steps is shown in Fig. 3. Table 1 presents attributes of the dataset.

4. Experimental Design, Materials and Methods

4.1. Data gathering

Dental health is a significant global concern, with various issues affecting populations in different regions. First, we tried to learn about various dental diseases, symptoms, the depth of damage to patients, and the widespread use of these dental issues. We found that dental diseases and treatments can be automated and AI can be detected. Therefore, we conducted an experiment with an experienced dental practitioner and collected X-rays from three different clinics under his supervision. We then photographed the X-rays using an Android mobile camera and created a collection. Finally, we decided that these X-rays could be used to classify and identify dental disorders by applying computer vision and deep learning algorithms [2].

We obtained 232 panoramic dental radiographs from dental clinics for this study. The dataset did not include teeth from individuals aged less than 10 years. The tooth structures in adults and

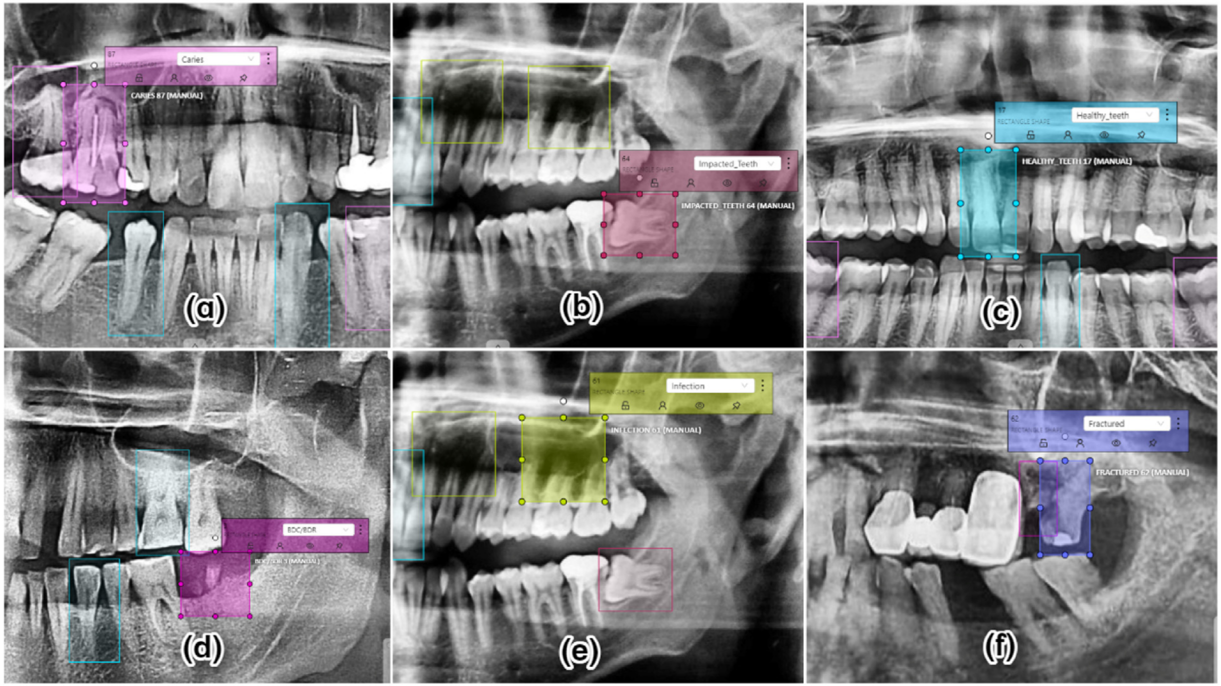

Fig. 1. Example of a) Caries, b) Impacted Teeth, c) Healthy Teeth, d) BDC/BDR, e) Infection, f) Fractured Teeth.

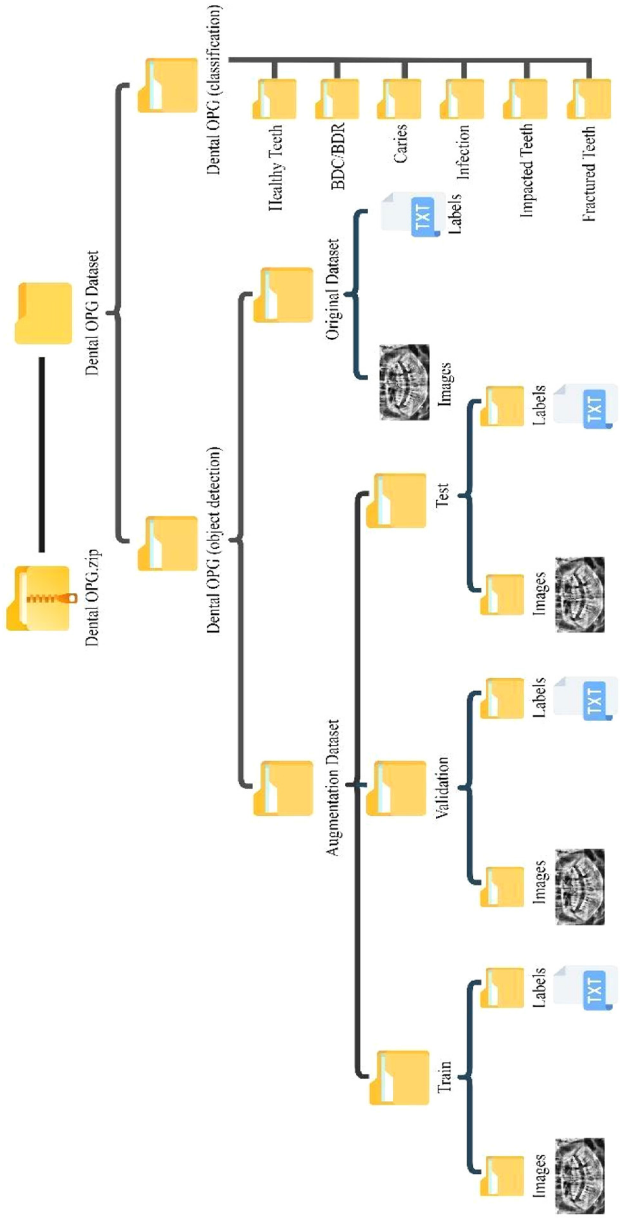

Fig. 2. Folder structure of the dataset.

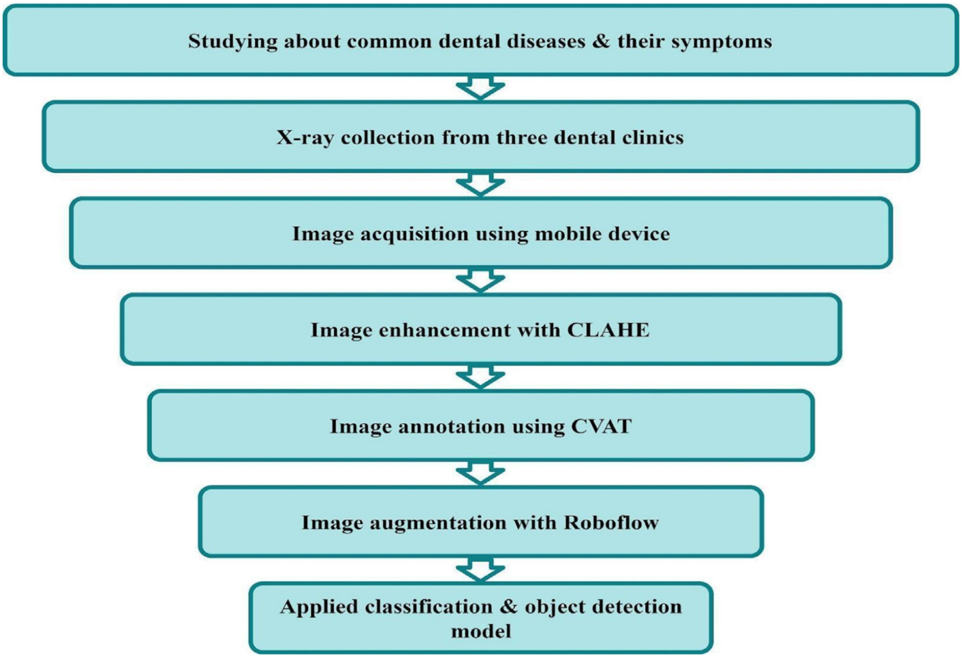

Fig. 3. Workflow of this study.

children differ. Therefore, this specific group of teeth has not been considered to maintain consistency across different age ranges. The dataset included six different dental diagnoses: caries, impacted teeth, infections, fractured teeth, broken-down crown/broken-down root (BDC/BDR), and healthy teeth. We stored the images in the JPG format and the labels in the JSON format. The dataset had two folders: object detection data and classification data. The object detection folder contains 232 raw images and labels and 604 augmented images, which are split into three files: train, test, and valid. The training folder has 558 images and labels; the test and valid folders both have 23 images and labels. On the other hand, classification folder contains six distinct classes as folders. The aspect ratio of these images was 16:9, and the median image ratio was 1176 × 603 pixels. The image resolution is 64 megapixels. The total number of instances across all classes was 1204. Each radiograph underwent an annotation process where regions corresponding to the identified diagnoses were labeled following a standard data annotation protocol. Thus, a fully anonymized dental panoramic image dataset was obtained, and no personal information could be used to identify patients. A batch of collected radiographs is shown in Fig. 4.

4.2. Image acquisition unit

The dataset was created using a Samsung A52 phone camera, specifically, the SM-A525F model. This camera captures images with a resolution of 64 megapixels and an f/1.8 aperture, allowing for clear and detailed photos. The camera has several helpful features, including an automatic exposure time ranging from 1/6000 s to 10 s. The ISO speed, which controls the camera's sensitivity to light, can be set automatically or manually with options ranging from 50 to 3200. The focal length of the camera was 26 mm, providing a balanced field-of-view for capturing the images. The photos were taken without using a flash, and a center-weighted average metering mode was used to ensure that the exposure was just right, meaning that the camera

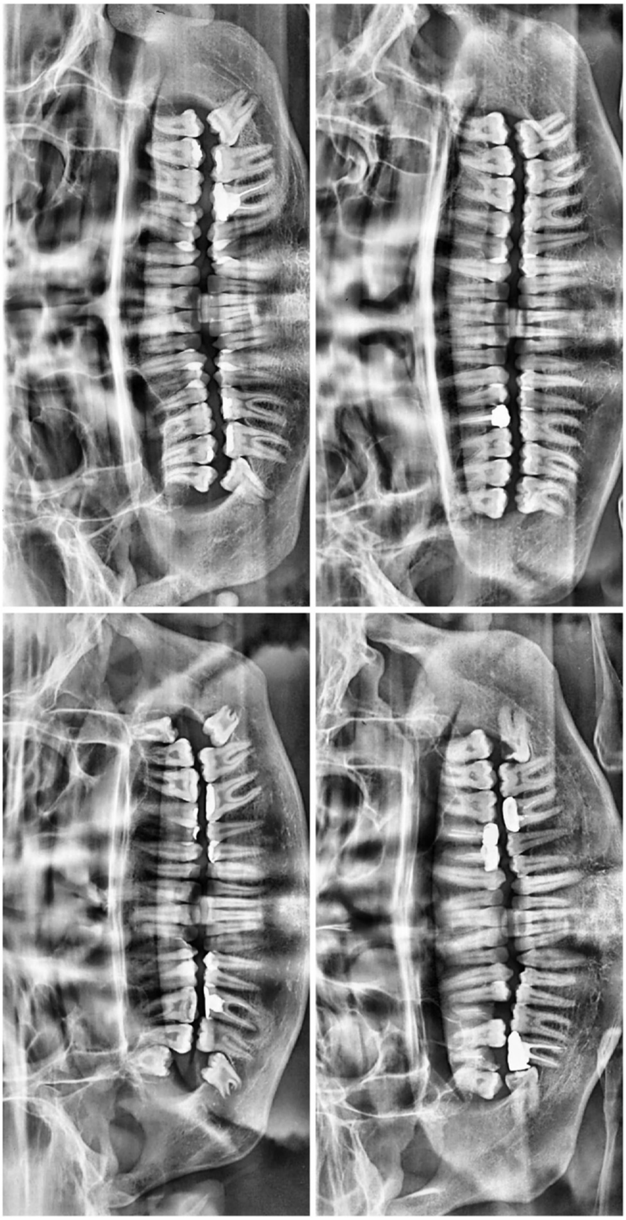

Fig. 4. Sample of collected radiographs.

Table 2
Brief description of the camera device.

Serial No.	Particulars	Description
1	Camera Maker	**Samsung A52**
2	Camera Model	**SM-A525F**
3	Resolution	**64 megapixels**
4	F-stop	**f/1.8**
5	Exposure time	**Auto, 1/6000 - 10s**
6	ISO speed	**ISO Auto, man: 50–3200**
7	Exposure bias	**0 step**
8	Focal length	**26 mm**
9	Flash mode	**No Flash**
10	Metering mode	**Centre weighted average**

focused on the central part of the image for light measurement. There was no exposure bias, which ensured that the images were consistent in quality. The pictures were taken under good lighting conditions, and the camera was handheld by a person, ensuring that the images were clear and of high quality. A detailed description of the camera device is provided in Table 2.

4.3. Image enhancement

Contrast-limited adaptive histogram equalization (CLAHE) is a technique used to enhance image contrast by targeting significant intensities within localized areas around each pixel. In our study, we applied CLAHE to improve the clarity and visibility of our images. In Fig. 5, the enhanced image demonstrates noticeably improved clarity and detail, making it easier to analyze compared to the original.

4.4. Image annotation

The CVAT annotation tool was used to annotate the panoramic photos. According to the bounding box, a line is stored in the text file. Consequently, there is one line for each annotation or bounding box. As this is a multiclass object detection study, we have six classes, ranging from 0 to 5. A label map file was used to convert this numeric value to an appropriate class level. To annotate images in the CVAT, we first created a new task and uploaded our images. Then, the annotation type (e.g., bounding box, square) was selected, and annotations were drawn on the images by clicking, dragging, or adding vertices. Subsequently, labels were assigned to the annotated objects, adjusted as needed, and navigated through the images to complete the annotations. CVAT automatically saves work and can be manually saved. Once done, we exported the annotations in desired format (e.g., YOLO) via the “Actions” dropdown, and downloaded the labels for use in the project. The annotation process was initially completed by a dentist with 16 years of experience using the annotation program CVAT, and it was later examined (addition, deletion, and confirmation) by a dentist with 32 years of experience. An example of placing a bounding box around problematic teeth is shown in Fig. 6.

4.5. Image augmentation

The dataset was processed using Roboflow and augmentation techniques were applied to enhance the training set. The preprocessing steps included auto orientation and resizing to fit within 640×640 pixels. The augmentation process generated three outputs per training example by applying horizontal flips, rotations between -15° and $+15^\circ$, shear transformations of

Fig. 5. Result of CLAHE implementation: a) Enhanced image, b) Raw image.

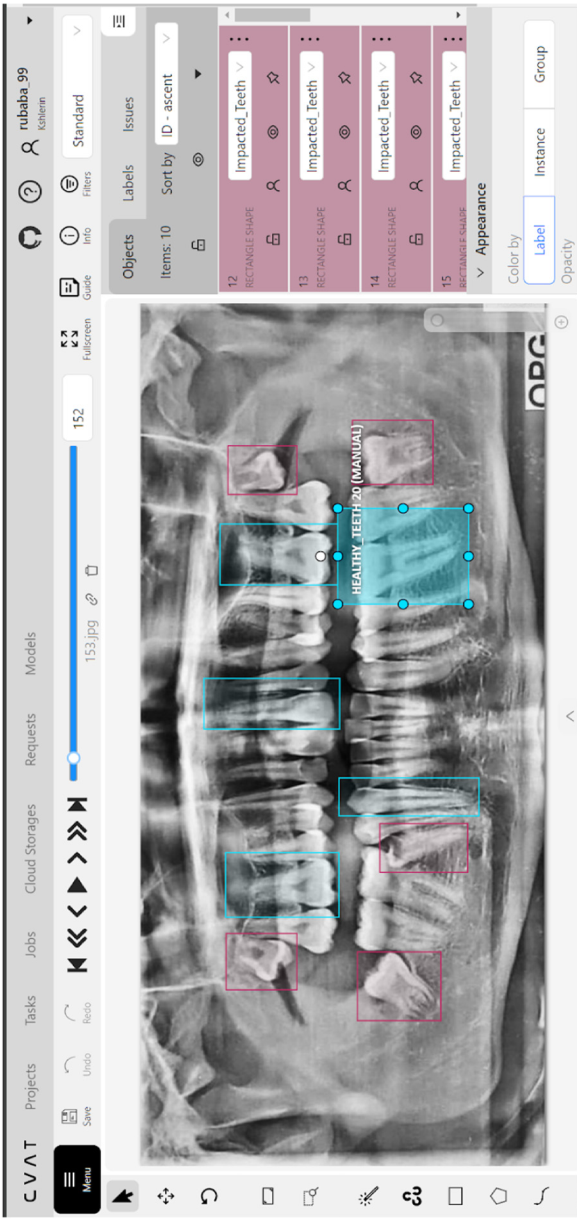

Fig. 6. Annotating using CVAT annotation tool.

Table 3
Summary of works related to dental diseases detection using deep learning.

Serial no	Ref.	Objectives	Datasets	Remarks
1	[4]	To identify dental issues among children	9562 images	Locally collected
2	[5]	To recognize dental conditions.	1488 OPG images	Collected from dental clinic
3	[6]	To develop automated diagnosis methods	929 raw X-ray images	Collected from hospital

Table 4
Dataset evaluation on Object detection algorithm.

Precision	Recall	mAP50
96.03	86.53	93.54

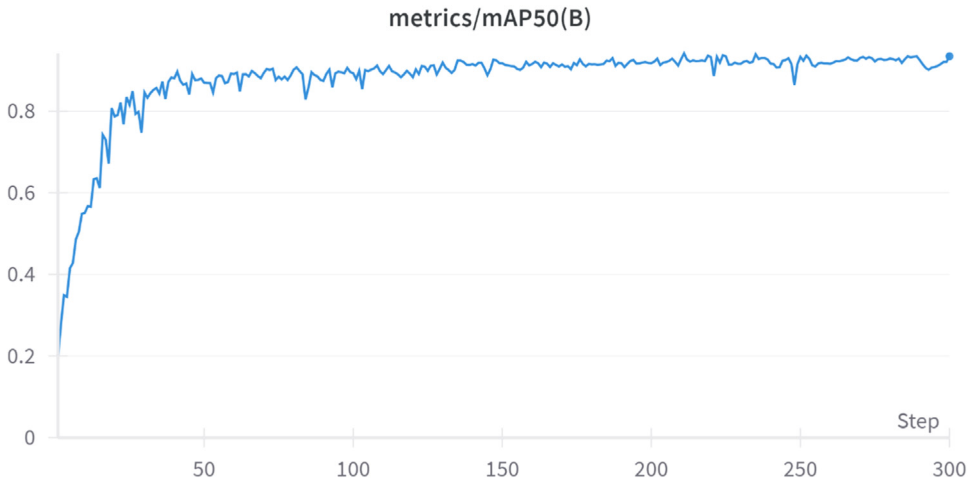

Fig. 7. Training graph of mAP50.

±10° both horizontally and vertically, and exposure adjustments within a range of −10 % to +10 %. In addition, mosaic augmentation [3] was applied, and the bounding box exposure was adjusted within the same exposure range, ensuring robust and varied training data for better model performance.

Table 3 summarizes previous research related to image processing in dental radiography. Developing such datasets in the healthcare domain, particularly in dental radiology, is inherently challenging because of the need for standardized, high-quality images and sensitivity of patient data. Consequently, many researchers have resorted to locally collecting images to advance their work. Our dataset fills this gap by providing valuable resources for machine learning and deep learning experts in dental informatics.

4.6. Dataset evaluation

To introduce our new "Dental OPG X-ray dataset," [1] we tested object detection performance using deep learning algorithm. Our detailed evaluation with pre-trained YOLOv7 model demonstrates strong detection capabilities and highlights the importance of this dataset. It shows that our dataset is a valuable benchmark. Table 4 illustrates the results of dataset evaluation and Fig. 7 presents the training graph of mAP50.

Limitations

First of all the dataset size was relatively small. Only 232 original images, which might limit the generalization of deep learning models to larger for more diverse populations. Second, the images were collected from only three clinics in Dhaka, Bangladesh; therefore, the dataset may not fully represent dental conditions from different regions or demographics. Additionally, the dataset excluded individuals under 10 years of age, which means that it cannot be used to study dental diseases in children. Finally, the images were captured using a single type of camera, which may introduce bias in the image quality and consistency.

Ethics Statement

All data collection procedures for this study were conducted in accordance with international ethical guidelines, including the Declaration of Helsinki and the International Ethical Guidelines for Health-related Research Involving Humans (CIOMS-WHO). Informed consent was obtained from the participating clinics and no personally identifiable information was included in the dataset. The study protocol, along with the informed consent forms, received approval from the Bangladesh Dental College & Hospital's local Ethics Committee (Protocol No: MK-198109).

Data Availability

Dental OPG XRAY Dataset (Original data) (Mendeley Data).

CRediT Author Statement

Rubaba Binte Rahman: Data curation, Formal analysis, Validation, Investigation, Methodology, Writing – original draft; **Sharia Arfin Tanim:** Conceptualization, Resources, Writing – original draft; **Nazia Alfaz:** Supervision, Investigation, Writing – review & editing; **Tahmid Enam Shrestha:** Investigation, Writing – review & editing; **Md Saef Ullah Miah:** Writing – review & editing, Project administration; **M.F. Mridha:** Writing – review & editing, Project administration.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.dib.2024.110970.

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